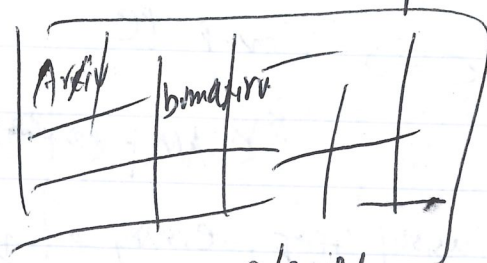


Data: composition of the pile by categories



evals approaches: { perplexity
testng.
instruction following.
scalp test-time compute: CoT
LM-as-judge: eval gen ~~tasks~~
Full-system: RAG/Agent

Data Curation: Data won't fall from the sky automatically

{ acquire data actively

=> pre-processing { transformation
filtering
removing
deduplication

Alignment: { follow ~~instruction~~ instructions
Tune style (format, length, tone, etc)
incorporate safety (refusal to answer questions)

step 1: SFT
step 2: HRLF

=> Verifiers

{ format = (code/math)
learned verifiers => LM-as-judge

RL algorithms:
 proximal-policy opt $\rightarrow$ PPO $\rightarrow$ from reinforcement learning
 direct-preference opt $\rightarrow$ GRPO: for preference data, simpler
 GRPO: remove value functions
 Group-Relative policy optimization

Overview of course logistics

Efficiency drive design

how to sequence the most compute from a given decision hardware $\rightarrow$ GPU

#1: Data preprocessing: avoid waste time on data cleaning

#2: Tokenization: working with raw bytes is promising but current

#3: Model Architecture: model variants to model architecture doesn't fit in

#4: Training/Modeling ~~efficiency~~: get away from one single epoch $\rightarrow$ reduce memory

#5: Scaling Laws/Interference: less compute on smaller model

#6: Alignment: human preference $\rightarrow$ desired for fine-tune/post-train
 tune model $\rightarrow$ If tune model to sequence case
 use smaller models

summary

scale away

design decision $\Leftarrow$ AI system

changing

Tomorrow: data-constrained